

SolidWorks Notes: Basics to Advanced

Introduction to SolidWorks

Overview of CAD, Part Modeling, Assembly Design, and Drawing Creation.

User Interface

FeatureManager Tree, Command Manager, Property Manager, Heads-Up Toolbar.

Sketching Fundamentals

Lines, Circles, Arcs, Rectangles, Slots, Smart Dimensions, Relations.

Part Modeling

Extrude, Revolve, Sweep, Loft, Rib, Shell, Draft, Hole Wizard.

Reference Geometry

Planes, Axes, Coordinate Systems, Reference Points.

Feature Operations

Fillet, Chamfer, Mirror, Pattern, Linear Pattern, Circular Pattern.

Configurations

Creating multiple design variations within a single part.

Assembly Design

Mates, Subassemblies, Interference Detection, Exploded Views.

Sheet Metal

Base Flange, Edge Flange, Hem, Jog, Flat Pattern.

Weldments

Structural Members, Cut Lists, Weld Beads.

Surface Modeling

Boundary Surface, Filled Surface, Knit Surface, Trim Surface.

Drawings

Views, Sections, Details, Dimensions, GD&T; Symbols.

Simulation Basics

Stress Analysis, Factor of Safety, Motion Study.

Rendering

PhotoView 360, Materials, Appearances, Lighting.

Industry Projects

Bracket, Coupling, Shaft, Gear, Vice Assembly, Pump Assembly.

Interview Questions

Extrude vs Revolve, Assembly Mates, Configurations, Sheet Metal Workflow.

8-Week SolidWorks Learning Plan

- Week 1: Sketching and Constraints
- Week 2: Basic Part Modeling
- Week 3: Advanced Features and Patterns
- Week 4: Assemblies and Mates
- Week 5: Engineering Drawings
- Week 6: Sheet Metal and Weldments
- Week 7: Surface Modeling and Simulation
- Week 8: Complete Mechanical Design Project